

Negative Exponents with Variable Bases

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	3
Lesson	Thread A · Day 2 — Negative Exponents with Variable Bases	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.AR.1.1

2. Standards

MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

Must. Integer exponents on monomial bases, negatives included: rewrite so that only positive exponents are left, and combine several laws in one expression.

Must not. Variables are nonzero wherever a negative exponent appears. No rational exponents, and monomials only.

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes I — 3.04's halving table rewritten with a letter, so $x^{-1} = 1/x$ is the next line of a pattern the class already owns.
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Board 4 — two negative exponents on two different letters; students say how many factors end up under the bar before they write anything.
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	Board 6 — when the room splits over whether the 2 gets squared, the sixty-second re-vote before the reveal.

4. Learning Target and Essential Question

Learning target. I can rewrite an algebraic expression with negative exponents using only positive exponents, and apply several laws of exponents to a monomial expression.

Essential question. What does a negative exponent on a variable mean, and how do several laws combine in one expression?

Building on. MA.8.NSO.1.3 (3.04–3.05) — the negative exponent law with numerical bases; T-A1 — product, power and quotient laws with variable bases.

Working toward. 3.08–3.09 — scientific notation, where 10^{-n} carries the small numbers; MA.912.NSO.1.2 — rational exponents in Algebra 1.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the 'today' line and nothing else yet.	Copy the target.
Warm-Up	5	Two minutes silent, then reveal. One sentence of reteach per question at most.	Four retrieval questions, alone, no notes.
Notes	11	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	13	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	53		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (13 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
$3x^{-2}$ and $(3x)^{-2}$ are not the same expression. Write the fraction for each and say exactly what moved.	DOK 3
$x^{-5} \cdot x^n = x^2$ has one answer. Is there an n that makes the product x^0 , and what would that mean?	DOK 3
Is x^{-3} ever a negative number? Say when, and what has to be true about x .	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	x^{-4}	$\frac{1}{x^4}$	$-x^4$ — negative exponent read as a negative expression [B1G-M 8.AR.1.1 misconception: negative exponents]; $-4x$ — exponent multiplied into the base
2	$\frac{1}{y^{-5}}$	y^5	$1/y^5$ — the sign was changed but the power was not moved [8.AR.1.1: law applied in one direction only]
3	A model's scale factor is $3x^{-2}$. The model is 12 centimeters long. Rewrite the factor using only positive exponents.	$\frac{3}{x^2}$	$1/(3x^2)$ — the coefficient was moved with the variable [8.AR.1.1: a negative exponent applies only to its own base]; $-3x^2$ — read as a negative; anything built from 12 — the unneeded figure was used
4	$y^{-3}z^{-4}$	$\frac{1}{y^3z^4}$	$1/(yz)^7$ — exponents on different bases were added [8.AR.1.1: like bases only]; $-y^3z^4$ — read as a negative
5	$x^4 \cdot x^{-6}$ Which expression is equivalent?	$C — 1/x^2$	(A) x^{-2} read as the opposite of x^2 [B1G-M 8.AR.1.1 misconception: negative exponents]; (B) $4 \cdot (-6)$ — multiplied the exponents [power of a power confused with a product]; (D) $4 - (-6)$ — subtracted instead of adding a negative [product law: add the exponents]
6	$(2x^{-3})^2$ Which expression is equivalent?	$D — 4/x^6$	(A) $-3 + 2 = -1$ instead of $-3 \cdot 2$ [power of a power confused with a product of powers]; (B) the coefficient was moved below the bar with the x [8.AR.1.1: a negative exponent applies only to its own base]; (C) the coefficient was not squared [8.AR.1.1: power of a product]
7	$\frac{m^3}{m^7}$	$\frac{1}{m^4}$	$m^4 — 7 - 3$ instead of $3 - 7$ [quotient law: top minus bottom]; m^{-4} — correct but not in the positive-exponent form asked for
8	Find the value of n : $x^{-5} \cdot x^n = x^2$	$n = 7$	-3 — added the given exponents [8.AR.1.1: product law, solve $-5 + n = 2$]; -7 — sign error in the integer subtraction
9	How are x^3 and x^{-3} related? Explain, and give the value of each when $x = 2$.	x^{-3} is the reciprocal of x^3 . When $x = 2$: $x^3 = 8$ and $x^{-3} = 1/8$.	' $x^{-3} = -8$ ' or 'they are opposites' — negative exponent read as a negative [B1G-M 8.AR.1.1 misconception: negative exponents]; ' -6 ' — exponent multiplied into the base

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / IEP. Give the fraction bar as a picture: a factor with a negative exponent crosses the bar and the minus disappears. The first four boards are one crossing each, and the student draws the arrow before writing the answer.

ELL. Simplified form has a specific meaning here. Write it on the board as a checklist — every exponent positive, every like base combined — and point at it before each board goes up.

Enrichment. Ask for $(2x^{-1}y^2)^{-3}$ with positive exponents only, and for the n that makes $x^{-2} \cdot x^n = 1$.

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the student has to name x^{-3} as the reciprocal of x^3 and give both values at $x = 2$. Read the boards, not the papers.

Vocabulary. **negative exponent law**; **simplified form**

Materials. Whiteboards and markers. Calculators are allowed on everything; $4^5 = 1,024$ in Example 2 is the one computation worth a calculator.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Variables with negative exponents — 2NL; Multiply and divide powers: variable bases — 85P.